



Laxmi Charitable Trust's
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Department of Information Technology (Bsc.IT Semester III)

Python Programming Practical

Practical - 4

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Practical No. 4 - Write Algorithm, Code and Output

Q. 1. Write algorithm for the following:

a. Write a program to perform slicing, basic and advanced indexing on NumPy arrays.

Code:

```
import numpy as np

arr1 = np.array([10, 20, 30, 40, 50, 60, 70])
```

```
arr2 = np.array([[10, 20, 30], [40, 50, 60], [70, 80, 90]])
print("Original 1D Array:", arr1)
print("Original 2D Array:", arr2)

print("\n----- Basic Indexing -----")
print("Element at index 2:", arr1[2])
print("Last element:", arr1[-1])
print("Element at (1,2) of 2D array:", arr2[1, 2])

print("\n----- Slicing -----")
print("Elements from index 1 to 4:", arr1[1:5])
print("Every second element:", arr1[::2])
print("Reversed array:", arr1[::-1])
print("Slice for 2D (Rows 0 to 1, All Columns):")
print(arr2[0:2, :])
print("Sliced(col 1):", arr2[:, 1])
print("\n----- Slicing -----")

print("Sliced(rows 0-1):\n", arr2[0:2, :])
```

```
print("\n----- Advanced Indexing -----")

indices = [0, 2, 5]
print("Elements at indices 0, 2, 5:", arr1[indices])

rows = [0, 2]
cols = [1, 2]
print("Advanced Indexing(0,2)&(1,2):", arr2[rows, cols])

print("Himanshu S037")
```

Output:

```
Original 1D Array: [10 20 30 40 50 60 70]
```

```
Original 2D Array: [[10 20 30]
```

```
[40 50 60]
```

```
[70 80 90]]
```

```
----- Basic Indexing -----
```

```
Element at index 2: 30
```

```
Last element: 70
```

```
Element at (1,2) of 2D array: 60
```

```
----- Slicing -----
```

```
Elements from index 1 to 4: [20 30 40 50]
```

```
Every second element: [10 30 50 70]
```

```
Reversed array: [70 60 50 40 30 20 10]
```

```
Slice for 2D (Rows 0 to 1, All Columns):
```

```
[[10 20 30]
```

```
[40 50 60]]
```

```
Sliced(col 1): [20 50 80]
```

```
----- Slicing -----
```

```
Sliced(rows 0-1):
```

```
[[10 20 30]
```

```
[40 50 60]]
```

```
----- Advanced Indexing -----
```

```
Elements at indices 0, 2, 5: [10 30 60]
```

```
Advanced Indexing(0,2)&(1,2): [20 90]
```

```
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```

b. Write a program to analyse dimensions and attributes of arrays.

Code:

```
import numpy as np

arr=np.array([[1,2,3], [4,5,6]])

print("Array:\n", arr)
print("Shape:",arr.shape)
print("Dimensions:", arr.ndim)
print("Size:", arr.size)
print("Data Type:",arr.dtype)
print("Item size:", arr.itemsize)
print("Total Bytes:", arr.nbytes)

print("Himanshu S037")
```

Output:

```
Array:  
[[1 2 3]  
 [4 5 6]]  
Shape: (2, 3)  
Dimensions: 2  
Size: 6  
Data Type: int64  
Item size: 8  
Total Bytes: 48  
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```

- Q. 2. What is the difference between basic slicing and advanced indexing?
- Q. 3. List and explain the common attributes of an array with examples.
- Q.4. Explain 1D, 2D and 3D arrays with examples using NumPy.